

CM0081 Formal Languages and Automata

Introduction to Haskell

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Features/Advantages

- Purely functional (verification)
- Statically typed (type-safe and maintainability)
- Lazy evaluation (unbounded data structures and performance)
- Garbage collected memory (no need for pointers)

Functions

Example (function application)

Factorial function.

```
fac n = product [1..n]
```

Note: We use “f n” instead of “f(n)” for function application.

Types

Question

Is the fac function correct?

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Example

```
import Numeric.Natural

fac :: Natural -> Natural
fac n = product [1..n]
```

By writing down the **type** of the function we could avoid run-time errors.

Types

Question

Is the fac function correct?

Example

```
import Numeric.Natural

fac :: Natural -> Natural
fac n = product [1..n]
```

By writing down the **type** of the function we could avoid run-time errors.

Other implementations for the factorial

Google for “The evolution of a HASKELL programmer”.

Curryfication

Example

Whiteboard.

Lists

Inductive definition

HASKELL has **built-in** syntax for lists, where a list is either:

- the empty list, written `[]`, or
- a first element `x` and a list `xs`, written `(x : xs)`.

Lists

Example (pattern matching on lists)

```
length :: [Int] -> Int
length []      = 0
length (x : xs) = 1 + length xs
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Question

Can we avoid the boilerplate listing? **Yes!**

Parametric Polymorphism

Example (basic functions from the `@Data.List@` library)

(i) Returns the length of a finite list as an `Int`.

```
length :: [a] -> Int
```

(ii) Append two lists.

```
(++) :: [a] -> [a] -> [a]
```

(iii) Extract the first element of a list, which must be non-empty.

```
head :: [a] -> a
```

Parametric Polymorphism

Example (basic functions from the `@Data.List@` library)

- (i) Extract the last element of a list, which must be finite and non-empty.

```
last :: [a] -> a
```

- (ii) Extract the elements after the head of a list, which must be non-empty.

```
tail :: [a] -> [a]
```

- (iii) Return all the elements of a list except the last one. The list must be non-empty.

```
init :: [a] -> [a]
```

- (iv) Test whether a list is empty.

```
null :: [a] -> Bool
```

Lazy Evaluation

Description

Nothing is evaluated until necessary.

Lazy Evaluation

Example

Infinite (unbounded) list.

```
ones :: [Int]
ones = 1 : ones
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The expression `take n` applied to a list `xs` returns the prefix of `xs` of length `n`, or `xs` itself if `n > length xs`.

```
take :: Int -> [a] -> [a]
```

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Which is the value of `take 5 ones`?

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```

Question

Which is the value of `take 5 ones`? `[1,1,1,1,1]`

Lazy Evaluation

Example (also in other programming languages)

Non-terminating function.

```
foo :: Int -> Bool
foo n = foo (n + 1)
```

Boolean disjunction.

```
bar :: Int -> Bool
bar n = True || foo n
```

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Question

Which is the value of bar 10?

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Question

Which is the value of bar 10? True

Lazy Evaluation

Example

(From stackoverflow.com/questions/30688558/)

```
dh :: Int -> Int -> (Int, Int)
dh d q = (2^d, q^d)

a, b :: (Int, Int)
a = dh 2 (fst b)
b = dh 3 (fst a)
```

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Question

Which is the value of a? (4,64)

Higher-Order Functions

Description

Functions are **first-class citizen**.

Higher-Order Functions

Example

The expression `map f xs` is the list obtained by applying `f` to each element of `xs`:

$$\text{map } f \ [x_1, x_2, \dots, x_n] = [f \ x_1, f \ x_2, \dots, f \ x_n]$$

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map :: (a -> b) -> [a] -> [b]
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Question

Which is the value of `map (+1) [1..5]`? `[2,3,4,5,6]`

Higher-Order Functions

Example

The function `foldr` applied to a binary operator, a starting value (typically the right-identity of the operator), and a list, reduces the list using the binary operator, from right to left:

$$\text{foldr } f \ z \ [x_1, x_2, \dots, x_n] = x_1 \ `f` \ (x_2 \ `f` \ \dots \ (x_n \ `f` \ z) \dots)$$

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Algebraic Data Types

Example

```
data Bool = True | False
```

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```

Functions by pattern matching

```
(||) :: Bool -> Bool -> Bool  
True  || _ = True  
False || x = x
```


Algebraic Data Types

Example (recursive data type)

```
data Nat = Zero | Succ Nat
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Algebraic Data Types

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```
data Nat = Zero | Succ Nat
```

Structural recursive function by pattern matching

```
(+) :: Nat -> Nat -> Nat  
Zero    + n = n  
(Succ m) + n = Succ (m + n)
```

Algebraic Data Types

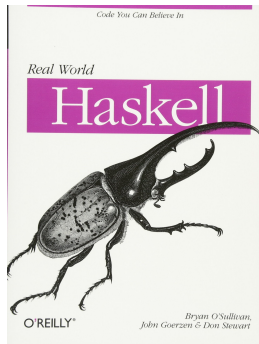
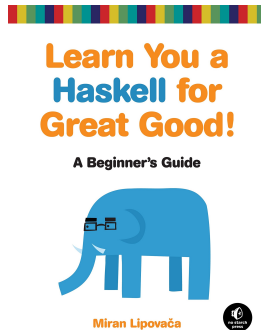
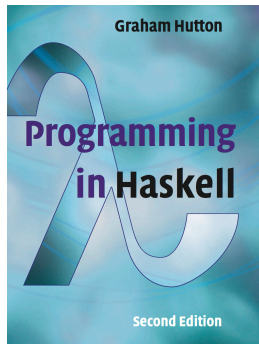
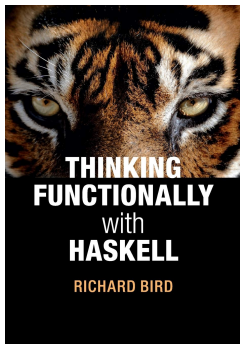
Example (polymorphic data type)

```
data List a = Nil | Cons a (List a)
data [] a   = []   | a : [a]
```

- HASKELL in Industry (www.haskell.org/haskellwiki/Haskell_in_industry).
- Applications (www.haskell.org/haskellwiki/Libraries_and_tools).

- Homepage: www.haskell.org
- GHC: The Glorious Glasgow Haskell Compilation System
- GHCi: Interactive interpreter
- Toolchain: www.haskell.org/downloads
 - For installing GHC we suggest to use GHCUP.
 - For installing libraries and compiling programs you can use STACK or CABAL-INSTALL.
- Hackage: The HASKELL package repository
- Community: www.haskell.org/community/

Some Books



Some Books

- Richard Bird (2015). Thinking Functionally with Haskell. Cambridge University Press.
- Graham Hutton [2007] (2016). Programming in Haskell. 2nd ed. Cambridge University Press.
- Miran Lipovača (2011). Learn You a Haskell for Great Good! No Starch Press.
- Bryan O'Sullivan, John Goerzen and Don Stewart (2008). Real World Haskell. O'Really Media, Inc.

Bonus Slides: Testing with QUICKCHECK

A paper

Claessen, Koen and Hughes, John [2000]. QuickCheck: A Lightweight Tool for Random Testing of Haskell Programs. ICFP'00. DOI: <https://doi.org/10.1145/357766.351266>.

[†]See www.sigplan.org/Awards/ICFP/.

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Most Influential ICFP Paper Award 2010[†]

The techniques described in the paper have spawned a significant body of follow-on work in test case generation. They have also been adapted to other languages

...

[†]See www.sigplan.org/Awards/ICFP/.

Bonus Slides: Testing with QUICKCHECK

An open source library

QUICKCHECK on Hackage.[†]

[†]<http://hackage.haskell.org/package/QuickCheck>.

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An open source library

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Commercialisation

QuviQ (www.quviq.com/).

[†]<http://hackage.haskell.org/package/QuickCheck>.

Bonus Slides: Testing with QUICKCHECK

Adaptations

QUICKCHECK has been ported to various languages (Wikipedia 2024-02-02).

C	C#	C++	CHICKEN	CLOJURE
COMMON LISP	CoQ	D	ELM	ELIXIR
ERLANG	F#	FACTOR	Go	Io
JAVA	JAVASCRIPT	JULIA	LOGTALK	LUA
MATHEMATICA	OBJECTIVE-C	OCAML	PERL	PROLOG
PHP	PONY	PYTHON	R	RACKET
RUBY	RUST	SCALA	SCHEME	SMALLTALK
STANDARD ML	SWIFT	TYPESCRIPT	VB.NET	VHILEY

Bonus Slides: Testing with QUICKCHECK

False positive

The program works properly but the test pointed out a fail:

- There is a bug elsewhere.
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False negative

There is a bug in the program but the test passed.

Recall Dijkstra's 1969 famous quote:

Testing shows the presence, not the absence of bugs.

Bonus Slides: Testing with QUICKCHECK

Example

See demo.